

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	<u>V.Naga Nischay</u>	Reg. No.:	192472255
Section / Batch:	CSE(AI)	Date:	28-08-26

Code of Conduct

I V.Naga Nischay (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Naga Nischay

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence
 - 2 questions on Conversational Agents and Dialog Acts
 - 2 questions on Language Generation
 - 2 questions on Machine Translation
 - 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:
Communication lacks clarity or confidence

Level 1:
Unable to communicate effectively

Confidence (15)

Level 4:
Displays high confidence, positive attitude and professional behavior throughout

Level 3:
Shows good confidence with minor issues

Level 2:
Low confidence and inconsistent behavior

Level 1:
Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

Student Response:

Q2:
Question Asked:
Student Response:

Q3:
Question Asked:
Student Response:

Q4:
Question Asked:
Student Response:

Q5:
Question Asked:
Student Response:

Q6:
Question Asked:
Student Response:

Q7:
Question Asked:
Student Response:

Q8:
Question Asked:
Student Response:

Q9:
Question Asked:
Student Response:

Q10:
Question Asked:
Student Response:

RUBRIC EVALUATION

Criterion	Weightage Assigned	Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

MOCK INTERVIEW ASSESSMENT REPORT

Student Name: Nischay

Register Number: 192472255

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

What is discourse in NLP? Explain the difference between a coherent discourse and a collection of unrelated sentences. Give a simple example.

Student Response:

Discourse in NLP is a group of connected sentences that convey a meaningful idea.

A coherent discourse has a logical flow, while unrelated sentences have no meaningful connection.

Q2:

Question Asked:

Consider the following discourse:

“Meena gave Kavya a laptop because she needed one for her project. She thanked her and started using it.”

Identify the likely antecedents of “she,” “her,” and “it.” Explain what constraints you would use to resolve these references, such as gender, number, recency, and semantic compatibility.

Student Response:

“She” most likely refers to Meena, “her” refers to Kavya, and “it” refers to the laptop.

The main constraints are gender, number, recency, and semantic compatibility to select the most suitable antecedent.

Q3:

Question Asked:

Imagine you are designing a customer-support chatbot.

A user says: “My internet connection is not working. Can you help me fix it?”

1. What dialog act(s) would you assign to this utterance?
2. How should the conversational agent interpret the user's intent?

Student Response:

1. Dialog act: It is a complaint/problem statement and a request for help.
2. Intent: The user wants to troubleshoot and fix their internet connection.

Q4:

Question Asked:

Suppose an NLP system has already determined:

Weather = Rainy

Temperature = 22°C

Recommendation = Carry an umbrella

Explain the main stages of Natural Language Generation (NLG) needed to convert this structured information into a natural sentence. Then give one suitable generated sentence.

Student Response:

The main stages of NLG are content planning, sentence planning, and surface realization.

For example: “It is rainy and 22°C, so you should carry an umbrella.”

Q5:

Question Asked:

Explain the difference between Transfer-based Machine Translation and Interlingua-based Machine Translation. Also mention one advantage and one limitation of each approach.

Student Response:

Transfer-based MT translates from source to target language using an intermediate language-specific transfer representation; it is easier to implement but needs separate transfer rules for language pairs.

Q6:

Question Asked:

Consider this conversation:

A: “Did you submit the assignment?”

B: “I finished writing it last night, but I still need to upload it.”

Analyze the response from a discourse/coherence perspective.

Student Response:

1. B's response is coherent because it directly answers A's question about the assignment.
2. The relationship is contrast: B finished writing it, but still needs to upload it.
3. The context tells us “it” refers to the assignment, because A had already mentioned the assignment.

Q7:

Question Asked:

A user tells a virtual assistant:

“Book me a flight to Delhi tomorrow morning.”

Assistant: “Sure. What is your preferred departure airport?”

Explain:

1. What dialog act is represented by the user's utterance?
2. What dialog act is represented by the assistant's response?
3. Why is the assistant's question necessary for completing the task?
4. What information should the dialogue system maintain in its dialog state after these two turns?

Student Response:

1. User's dialog act: A request to book a flight to Delhi tomorrow morning.
2. Assistant's dialog act: A clarification question requesting the departure airport.
3. The airport is necessary because the system needs the departure location to search and book the correct flight.
4. The dialog state should maintain: destination = Delhi, date = tomorrow, time = morning, task = flight booking, departure airport = not yet provided.

Q8:

Question Asked:

A dialogue system has decided to communicate:

Intent = Inform the user that their flight is delayed.

Reason = Bad weather

Delay = 3 hours

Explain how the surface realization stage converts this structured meaning into a grammatically correct natural-language sentence. Then provide one suitable generated sentence.

Student Response:

Surface realization converts the structured meaning into a grammatically correct sentence by choosing the right words, grammar, and sentence structure.

Example: "Your flight is delayed by 3 hours due to bad weather."

Q9:

Question Asked:

Consider a statistical machine translation system translating:

English: "The cat is on the mat."

Tamil: "பூனை பாயின் மேல் உள்ளது."

Explain how a statistical MT system would choose the most probable Tamil translation. Mention the roles of the translation model and language model, and explain why the system may choose one translation over another.

Student Response:

The translation model gives the probability of possible English-to-Tamil word/phrase translations, while the language model checks which Tamil sentence is more natural and grammatically correct.

The system chooses the translation with the highest overall probability, so it may prefer "பூனை பாயின் மேல் உள்ளது" because it is both a likely translation and a natural Tamil sentence.

Q10:

Question Asked:

You are designing a conversational travel assistant.

User: "I want to travel to Chennai tomorrow."

Assistant: "Sure. How would you like to travel?"

User: "By train. I prefer an evening journey."

The system must now generate an appropriate response.

Explain:

1. What information should the system extract and maintain in its dialog state?
2. What dialog acts are present in the user's two responses? Give one example of the final generated response.

Student Response:

1. Dialog state: Destination = Chennai, date = tomorrow, transport = train, preferred time = evening.
2. Dialog acts: The user first makes a travel request, then provides information and preference .Example: "Sure, I'll look for trains to Chennai tomorrow evening."



Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	4	4	4	4	4	16	100
2	4	4	4	4	4	16	
3	4	4	4	4	4	16	
4	4	4	4	4	4	16	
5	4	4	4	4	4	16	
6	4	4	4	4	4	16	
7	4	4	4	4	4	16	
8	4	4	4	4	4	16	
9	4	4	4	4	4	16	
10	4	4	4	4	4	16	

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____